A Wearable Smart Glove and Its Application of Pose and Gesture Detection to Sign Language Classification

Problem Addressed

Communication through sign language requires accurate recognition of hand poses and dynamic gestures. Traditional vision-based sign language recognition systems rely on cameras and image processing, which are sensitive to lighting conditions, background complexity, and camera positioning. Therefore, a more reliable wearable system is needed to capture hand motion and gestures directly from the user.

Research Gap

Many existing smart glove systems require complex fabrication, expensive hardware, or external computers for processing. In addition, some systems focus only on detecting hand poses but cannot recognize dynamic gestures. These limitations reduce the practicality of wearable sign language recognition systems.

Key Contribution

The research proposes a **wearable smart glove system** that can detect hand poses and gestures and classify sign language in real time. The glove integrates strain sensors, an accelerometer, and a microcontroller to perform on-board machine learning classification.

Methodology

The proposed system consists of the following components:

- A **conductive knitted glove** that acts as strain sensors for detecting finger bending
- **16 resistive strain sensors** placed across hand joints
- A **3-axis accelerometer** to capture hand orientation and motion
- An **STM32 microcontroller** that performs signal processing and gesture classification
- A **Long Short-Term Memory (LSTM) neural network** used to classify gestures

Sensor data from the glove and accelerometer is processed and fed into the neural network to detect sign language gestures

Dataset / Gesture Vocabulary

The system was tested using **24 American Sign Language (ASL) gestures**, including:

- **11 static poses** (letters)
- **13 dynamic gestures** (words and phrases)

Examples include letters A–L and words such as Hello, Thank You, Please, Yes, and Eat

Results

The proposed system achieved high accuracy in recognizing sign language gestures:

- **Training accuracy:** ~99%
- **Testing accuracy:** ~96.3% for segmented gesture examples
- **Real-time streaming prediction accuracy:** about **91% correct gesture detection**

These results demonstrate that the system can effectively classify both static poses and dynamic gestures.

Significance

This research demonstrates that wearable smart gloves combined with machine learning can provide a practical solution for real-time sign language recognition. The system enables intuitive human–computer interaction and can help improve communication between deaf individuals and others.

Future Work

Future improvements suggested in the paper include:

- Testing the system with **multiple users** to improve generalization
- Increasing the **number of recognized gestures**
- Improving the **machine learning model and sensor placement**
- Adding additional sensing modalities for broader applications.